LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 5

Course Code: INT396	Unit: 5
Course Title: UNSUPERVISED LEARNING	Unit Title: Association Rule Mining and Anomaly Detection
Course Outcome: CO5	Unit Mapped RBT: L3 (Apply)
Paper: Sample Subjective Questions	Total Questions: 10
Maximum Marks: 10 x 10 = 100	Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply market basket analysis to the transaction records of a supermarket. (10 Marks)

(a) State the steps that take the raw receipts through to a set of association rules.

(b) Use two example rules to describe the actions the store could take as a result.

[L3 - Apply]

Q2. Compare support, confidence and lift as filters applied to a large set of candidate rules. (10 Marks)

(a) State what each of the three removes from the candidate set.

(b) Use them in order to describe how several thousand rules would be reduced to a usable handful.

[L3 - Apply]

Q3. Examine the effect of the minimum support threshold on an Apriori run. (10 Marks)

(a) State what the threshold controls and how it is expressed.

(b) Use this to describe what goes wrong when the threshold is set too high and when it is set too low.

[L3 - Apply]

Q4. Apply the Isolation Forest method to a fraud-screening problem. (10 Marks)

(a) State how the method assigns an anomaly score to a record.

(b) Use this to explain why it remains fast enough for a database of many millions of records.

[L3 - Apply]

Q5. Compare the Local Outlier Factor with a single global distance threshold as a way of finding unusual records. (10 Marks)

(a) State what each of the two compares in order to reach a decision.

(b) Use a dataset holding one dense region and one sparse region to show where the two disagree.

[L3 - Apply]

Q6. Use the distinction between anomaly detection and novelty detection to set up a monitoring system. (10 Marks)

(a) State the difference in the data that each of the two is given.

(b) Use this to state which of the two suits a newly installed machine with no history of faults, and why.

[L3 - Apply]

Q7. Apply the FP-Growth algorithm to a transaction database. (10 Marks)

(a) State what the FP-tree holds and how it is constructed from the transactions.

(b) Compare the number of database scans it needs with the number Apriori needs, and explain the difference.

[L3 - Apply]

Q8. Examine an association rule reported with a confidence of 0.92 and a lift of 1.01. (10 Marks)

(a) State what each of the two figures says about the rule.

(b) Use this to decide whether the rule is worth acting on, and justify the decision.

[L3 - Apply]

Q9. Demonstrate how an anomaly-detection system should be evaluated once it is in service. (10 Marks)

(a) State what should be recorded about every alert the system raises.

(b) Use those records to describe two figures you would report to management each month.

[L3 - Apply]

Q10. Relate the choice of anomaly threshold to the cost of the two kinds of error. (10 Marks)

(a) State what a false positive and a false negative each cost in a card fraud setting.

(b) Use this to describe how you would set the threshold for a bank, and what you would monitor afterwards.

[L3 - Apply]

----- End of Sample Question Set -----